

Project title

SmartGlycoAI: Intelligent Decision Support Platform for Personalized Management of Type 1 Diabetes.

Group members

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Problem description

Type 1 Diabetes Mellitus is a chronic disease that requires continuous blood glucose monitoring, constant dietary control, and rigorous insulin administration. Inadequate management of this condition can lead to severe episodes of hypoglycemia or hyperglycemia, as well as medium and long-term complications. The major current difficulty lies in the fact that the integrated interpretation of dietary intake, physical activity, and biometric data still relies heavily on the patient's own experience and judgment.

Target users - People suffering from Type 1 Diabetes.

Project goal

The main goal is to develop and validate a cross-platform application (Responsive Web and/or Mobile) that acts as an intelligent personal clinical decision support system. The proposed solution aims to integrate real-time biometric data (CGM sensors via Bluetooth) with a computer vision system that recognizes meals and estimates carbohydrates. The application will use adaptive Artificial Intelligence models to provide personalized recommendations regarding insulin doses, physical activity, and preventive glycemic risk alerts, without ever replacing a doctor's medical decision.

Competing applications

Positive aspects: They allow for continuous, real-time monitoring of blood glucose values.

Negative aspects: They fail by leaving the burden of interpreting all the integrated data (glucose, food, exercise) up to the user's intuition and empirical experience.

Our solution: SmartGlycoAI fixes this design flaw by automating the nutritional calculation (through computer vision to recognize food) and using Artificial Intelligence to turn raw data into proactive, precise, and personalized recommendations.